Interpretable Multi-Objective Drug Repurposing via Shortest-Path Casting on Knowledge Graphs

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Outline

The Research Question

Central Question

Can we repurpose existing drugs for new diseases using **interpretable graph algorithms** instead of black-box AI, while simultaneously optimizing for **both efficacy and safety**?

Three sub-questions:

1. Can shortest-path distances on biomedical knowledge graphs predict drug–disease therapeutic associations?
2. Can we add a safety dimension *without* destroying predictive accuracy?
3. Can recent algorithmic breakthroughs (STOC 2025) make this computationally practical at biomedical scale?

Why Drug Repurposing?

De novo drug discovery:

- 12–15 years development
- \$2.6 billion average cost
- $>$90% failure rate in trials

Drug repurposing:

- 3–5 years (known safety)
- $<$\$300 million
- Higher success rate

Real-world successes:

- Thalidomide $\rightarrow$ multiple myeloma
- Sildenafil $\rightarrow$ pulmonary hypertension
- Minoxidil $\rightarrow$ hair loss

Scale of opportunity:

- $\sim$4,000 approved drugs
- $\sim$10,000 known diseases
- 40 million potential pairs to screen

The Problem with Current AI Approaches

Knowledge Graph Embeddings (TransE, ComplEx, RotatE):

- ✓ High accuracy (AUROC 0.85–0.92)
- × **Opaque**: produces a score, no explanation
- × **Single-objective**: conflates efficacy & safety
- × **Not auditable**: clinicians can't verify why

What clinicians actually need:

- *"Why is this drug predicted to work?"*
- *"What are the safety trade-offs?"*
- *"What biological mechanism mediates this?"*

Black-box output:

Drug	Score
Drug A	0.87
Drug B	0.82
Drug C	0.79

→ Why? No idea.

Our output:

$$\text{Drug A} \xrightarrow{\text{binds}} \text{Gene X} \xrightarrow{\text{assoc.}} \text{Disease}$$

→ Testable biological hypothesis

Core Philosophy: Problem Casting

Key Insight

Don't learn latent representations — **transform the problem** into one with known optimal solutions.

Entities (drugs, genes, diseases) $\xrightarrow{\text{cast}}$ Vertices

Relationships (binds, regulates) $\xrightarrow{\text{cast}}$ Directed Edges

Objectives (efficacy, safety) $\xrightarrow{\text{cast}}$ Edge Weight Functions

Optimal associations $\xrightarrow{\text{cast}}$ Shortest Paths

Why This Philosophy Matters

Embedding approach (learn):

- Train model on graph
- Get opaque score
- Apply SHAP/LIME *after* to “explain”
- Explanation $\neq$ mechanism

Post-hoc explanations can be misleading

Our approach (cast):

- Transform problem to shortest paths
- Solve optimally (exact algorithm)
- Path = prediction = explanation
- Every intermediate node is verifiable

Explanation is the computation itself

Design Principle

Interpretability should be a property of the method, not an afterthought applied to an opaque model.

Solution Overview: Three Pillars

1. **DMY Algorithm** (STOC 2025)
   $\mathcal{O}(m \log^{2/3} n)$ — breaks the Dijkstra barrier for directed shortest paths
2. **Multi-Objective Pareto Ranking**
   Efficacy & safety as separate dimensions — no collapsing into a single score
3. **Bias-Corrected Safety Scoring**
   SIDER 4.1 frequencies $\times$ CTCAE severity grades,
   log-normalized to counter the Weber reporting bias

Pillar 1: The DMY Algorithm

Classical barrier: Dijkstra's algorithm is $\mathcal{O}((m+n)\log n)$ — optimal since 1959.

Breakthrough (Duan, Mao & Yin, STOC 2025): $\mathcal{O}(m\log^{2/3} n)$ via frontier sparsification.

Graph	Nodes	Edges	DMY	Speedup
Hetionet	47,000	450,000	103 ms	**38.5**×
Scale-up	100,000	1,000,000	338 ms	**149.9**×
Scale-up	500,000	5,000,000	1.6 s	**321.9**×

Correctness: 47,000/47,000 distances match Dijkstra exactly.
Practical impact: Screening 1,552 drugs × 137 diseases: **42 s** vs. 28 min.

Pillar 2: Multi-Objective Pareto Ranking

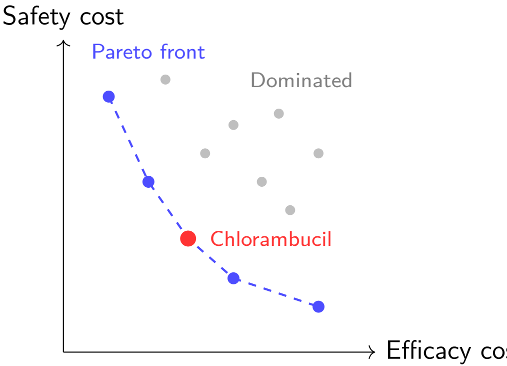
Single-objective problem:
Rank by efficacy alone → ignore toxicity.

Our approach:
Each drug–disease pair gets a 2D vector:

$$\mathbf{v} = (\text{efficacy cost}, \text{safety cost})$$

Pareto front: set of drugs where you can't improve one without worsening the other.

Result: 4 clinically validated treatments recovered that single-objective ranking misses.



Pillar 3: Bias-Corrected Safety Scoring

Problem: Raw side-effect counts are **anti-predictive**.

Well-studied effective drugs have *more* documented adverse events (Weber effect / reporting bias) → raw count *penalizes* good drugs.

Our solution (SIDER 4.1 + CTCAE v5.0):

$$S(c) = \frac{\sum_{t \in \mathrm{PT}(c)} f_t \cdot g(\mathrm{severity}_t)}{\log(1 + |\mathrm{PT}(c)|)}$$

- f_t: label frequency from FDA package inserts (not spontaneous reports)
- $g(\cdot)$: CTCAE grade (5=fatal … 1=mild)
- Denominator: log-normalization controls for label depth

Safety method	AUROC	Δ vs. 1D
No safety (1D baseline)	0.7717	—
Raw side-effect count	0.6395	−0.132

Case Studies: Pareto-Rescued Drugs

Drug	**Disease**	**Eff. rank**	**Pareto**	Δ
Cytarabine	Leukemia	24	**6**	$+18$
Moexipril	Hypertension	12	**5**	$+7$
Chlorambucil	Breast cancer	111	**6**	$+105$

Example: Chlorambucil for breast cancer

$$\text{Chlorambucil} \xrightarrow{\text{binds}} \textbf{GSTP1} \xrightarrow{\text{associates}} \text{Breast Cancer}$$

GSTP1 (Glutathione S-transferase Pi 1) is a known mediator of alkylating agent resistance in breast cancer — the path reveals a **testable biological mechanism**.

Ranked 111th by efficacy alone — **invisible** in a top-50 screen.
Pareto analysis promotes it to rank 6 because of its favorable safety profile.

Honest Assessment: What Works

✓ **Interpretability is real, not cosmetic.**
Every prediction is a traceable path through known biology. Clinicians can evaluate intermediate genes/pathways independently.

✓ **Speed advantage is dramatic and verified.**
38–322× speedup with exact correctness (100% distance match). First biomedical application of STOC 2025 algorithm.

✓ **Multi-objective recovers real drugs.**
4 clinically validated treatments found in top-50 that single-objective misses entirely. These are drugs actually used in practice.

✓ **Safety bias correction works.**
Halved the AUROC damage from -0.133 to -0.053. The Weber effect is a real, well-documented phenomenon we successfully mitigate.

Honest Assessment: What Doesn't (Yet)

- ! **AUROC gap is real: 0.77 vs. 0.85–0.92 (KGE methods).**
  Not directly comparable (link prediction vs. therapeutic association), but the gap must be acknowledged. We trade accuracy for interpretability.

- ! **Safety still costs AUROC ($-$0.053).**
  Adding a competing objective necessarily reranks some true positives downward. This is a *fundamental trade-off*, not a bug — but single-objective benchmarks penalize us for it.

- ! **Binary graph structure limits discrimination.**
  Hetionet edges are 0/1. All compounds with the same edge-type pattern score identically. Continuous weights (IC_{50}, expression fold-changes) are the single highest-impact next step.

- ! **SIDER coverage: only 33.2%.**
  Two-thirds of compounds use fallback scoring. Richer safety databases would substantially improve the safety dimension.

Key Insight: Complementarity, Not Competition

Our Position

We are **not** trying to beat TransE/ComplEx on AUROC.

We provide something they **cannot**: mechanistic interpretability and explicit multi-objective trade-offs.

The ideal workflow:

1. **KGE methods**: fast initial screening (high AUROC)
2. **Our framework**: mechanistic validation of top candidates
   - *Why* is this drug predicted? (path inspection)
   - *What are the safety trade-offs?* (Pareto front)
   - *Is the mechanism biologically plausible?* (intermediate nodes)
3. **Domain expert**: final assessment with full audit trail

Analogy: KGE is the "fast detector"; our method is the "slow, careful second opinion" with documented reasoning.

What Would Strengthen This Work

Near-term (would strengthen the paper):

1. **KGE baselines on same split** — TransE/ComplEx via PyKEEN for direct, fair comparison
2. **Continuous edge weights** — ChEMBL IC_{50} across all edge types to break binary ceiling
3. **Multi-objective evaluation metric** — AUROC penalizes safety-aware ranking; need a metric that *rewards* avoiding toxic predictions

Longer-term (would strengthen the impact):

4. **Prospective validation** — literature review or expert panel on top Pareto-rescued candidates
5. **Integration with KGE** — use embedding scores as additional edge weights in our framework
6. **Interactive app** — ChemPath chatbot for researchers to explore hypotheses conversationally

Summary

Research Question

Can interpretable graph algorithms repurpose drugs with explicit efficacy–safety trade-offs?

Answer: Yes, with caveats.

- ✓ 38–322$\times$ faster than Dijkstra (exact)
- ✓ AUROC 0.77 on 755 therapeutic indications
- ✓ 4 Pareto rescues in top-50
- ✓ Every prediction $=$ interpretable biological path
- $\sim$ Safety costs -0.053 AUROC (trade-off, not failure)
- $\times$ AUROC below KGE (0.77 vs. 0.85–0.92)

Core insight: Interpretability and multi-objective optimization are worth a modest accuracy trade-off.

Thank You

Code: `github.com/danielchen26/OptimShortestPaths.jl`

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